

Technical Report: Sanskrit-to-English Neural Machine Translation

1. Architecture Overview

We employed the pre-trained NLLB-200 (No Language Left Behind) Transformer model, which is suitable for translation between languages with low resources such as Sanskrit. In particular we used the `facebook/nllb-200-distilled-600M` distilled model.

It is based on the standard Transformer encoder-decoder architecture though it has been specially trained with 200+ languages and uses special tokens for Devanagari script (Sanskrit) and Latin script (English) to deal with the morphological richness of the source language.

2. Methodology

For the pre-processing step used the following AutoTokenizer: a Sanskrit (`san_Deva`) source and English (`eng_Latn`) target. Sentences were tokenized and a maximum length of 128 tokens was applied.

- Done batch inference (batch size=8) on test set. Introduced `forced_bos_token_id` to make sure the output is in the target language when forced generation is used.
- **Evaluation Metrics:**
BLEU Score is calculated by NLTK's `corpus_bleu` function with default geometric mean 1-4 grams and Method 1 smoothing.
BERTScore is computed using the `bert-score` library and the `roberta-large` model, and the `rescale_with_baseline=True` parameter to set F1 scores to be normalized.

- Efficiency: Total inference time (1000 sentences) and total trainable parameters.

3. Results

- **BLEU Score:** 10.97
- **BERTScore F1:** 0.4675 (Rescaled)
- **Inference Time (Total):** 96.20 seconds
- **Total Model Parameters:** 615,073,792

4. Discussion

The NLLB-200 model showed a good capacity in capturing the semantic structure of Sanskrit. The BLEU score (10.97) reflects the difficulties of low-resource NMT and the n-gram metrics' strict definition of an exact match, while the BERTScore provides a measure of meaningful semantic alignment. In qualitative examples, the model effectively managed various types of sentences, ranging from technical and informative texts to scriptural texts.

5. References

1. Costa-jussà, Marta R., et al., "No Language Left Behind: Scaling Human-Centered Machine Translation. Papineni, Kishore, et al. (1998). BLEU: a method for automatic evaluation of machine translation.
2. Zhang, Tianyi, et al. "BERTScore: Evaluating Text Generation with BERT.